Implementation of lexical analyzer using lex tool.

cdlab3.l

```
%{
#include <stdio.h>
%}

%%

"int"|"float"|"if"|"else"       { printf("Keyword: %s\n", yytext); }
[0-9]+                          { printf("Number: %s\n", yytext); }
[a-zA-Z_][a-zA-Z0-9_]*          { printf("Identifier: %s\n", yytext); }
"+"|"-"|"*"|"/"|"="             { printf("Operator: %s\n", yytext); }
";"|","|"(" | ")" | "{" | "}"   { printf("Symbol: %s\n", yytext); }
[ \t\n]+                        { /* ignore whitespace */ }
.                               { printf("Unknown: %s\n", yytext); }

%%

int main() {
   yylex();
   return 0;
}

int yywrap() {
   return 1;
}
```

OUTPUT:

```
kits@kits-Veriton-M200-H310:~/Desktop$ lex cdlab3.l
kits@kits-Veriton-M200-H310:~/Desktop$ cc lex.yy.c
kits@kits-Veriton-M200-H310:~/Desktop$ ./a.out
10
Number: 10
float
Keyword: float
kits
Identifier: kits
```

Write a lex program to count the number of words and number of lines in a given file or program.

cdlab8.l

```
%{
int ccnt=0,wcnt=0,lcnt=0;
%}
word [^ \t\n]+

%%
{word} {wcnt++;ccnt+=yyleng;}
\n  {lcnt++;}
. {ccnt++;}
%%
void main()
{
yylex();
printf("\n The character count =%d",ccnt);
printf("\n The word count = %d",wcnt);
printf("\n The line count = %d",lcnt);
printf("\n");
}
int yywrap()
{
return 1;
}
```

OUTPUT:

```
kits@kits-Veriton-M200-H310:~/Desktop$ lex cdlab8.l
kits@kits-Veriton-M200-H310:~/Desktop$ cc lex.yy.c
kits@kits-Veriton-M200-H310:~/Desktop$ ./a.out
cse
cd lab

 The character count =9
 The word count = 3
 The line count = 2
```

***Write a Lex program to check whether the given number is even or odd?**

eo.l:

```
%{
#include<stdio.h>
int i;
%}
%%

[0-9]+      {
                i=atoi(yytext);
                if(i%2==0)
                        printf("Even");
                else
                printf("Odd");
            }
%%
int yywrap()
{}
int main()
{
    yylex();
    return 0;
}
```

Output:

```
$ lex eo.l
$ cc lex.yy.c
$ ./a.out
22
Even
33
Odd
```

***Write a Lex code to count the frequency of the given word in a file?**

input.txt:

```
kits
jits
vits
kits
```

count.l

```
%{
#include<stdio.h>
#include<string.h>
char word [] = "kits";
int count = 0;
%}

%%
[a-zA-Z]+     { if(strcmp(yytext, word)==0)
```

```
                    count++; }
.  ;
%%
int yywrap()
{
    return 1;
}
int main()
{
        extern FILE *yyin, *yyout;
        yyin=fopen("input.txt", "r");
        yylex();
        printf("%d", count);
}
```

Output:

```
$ lex count.l
$ cc lex.yy.c
$ ./a.out
2
```

***Write a Lex Program to find the successor of the given String?**

succ.l:

```
%{
int i,l,m;
%}
%%
[a-z]*          {
          for(i=0;i<yyleng;i++)
          {
          l=yytext[i];
          if(l==(m='z'))
          l='a';
          else
          l++;
          printf("%c",l);
          }
          }
%%
int main()
{
    yylex();
    return 0;
}
int yywrap()
{}
```

Output:

```
$ lex succ.l
$ cc lex.yy.c
$ ./a.out
kits
ljut
```